

One of the security constraints of an embedded system in comparison to other normal software systems is its difficulty in procuring software updates which can fix known issues (This is called ‘intrusion containment’). Let’s elaborate a bit on this with a comparative example: Windows XP and below had a list of severe vulnerabilities exposed to the general public via the (in)famous Internet Explorer. This vulnerability was caused by two things – usage of ActiveX controls which allowed an unusual amount of control on the system by a remote attacker, and secondly IE’s deep penetration into the Windows OS. However, these issues associated with IE were resolved with patches and service packs which ensured that the users’ computers were protected against further threats. In due course of time, IE 7 was released as separate software for Windows XP and it filled the gaps left by IE 6 to quite an extent.

The two security issues here that stand out in terms of importance to embedded systems are:

1 Intrusion Containment: When an intrusion happens, the hack shouldn’t be able to corrupt systems other than what was compromised. In case of Internet Explorer, its depth of connectedness to Windows opened new loopholes and allowed attackers and malicious websites to read and write data or run malicious code on the system. If only IE were to be just another application installed on top of the OS with no deeper level integrations, the threats would have been smaller than they were. In case of embedded systems, it’s vital that the components are isolated from other components from the attack perspective and a compromise of one component (let’s say the communication component) shouldn’t affect other components (e.g. motor controls).



Flashing a phone for installing updates can be complicated even though they are not the most complicated or critical embedded systems

2 Ease of Update: It’s one thing to install an update for Windows and quite another to install one for an embedded system. The difference in level of ease (or difficulty) can be understood by comparing a Windows installation